

Surgery demonstration

Autoclave list:

- Surgical tools
- Aluminium foil
- Cotton swabs
- Drillpiece
- tips to apply UV glue and superbond
- stereotax adapter
- implantation rod for GRIN lens

Sterile field preparation (with regular gloves):

- Disinfect all equipment that will be touched.
- Prepare 2 petri dishes outside the sterile zone for chlorhexidine and PBS.
- 2 sterile drapes on either side of the stereotax (handle only the edges).
- Place autoclaved tools, cotton swabs, insulin syringe needles, a 27 G needle and aluminium foil in the middle of the drape without letting the outside of the autoclave bag touch the sterile drape.

Injection pipette preparation

- Fill the injection pipette with mineral oil using a syringe before mounting on the stereotax.
- Mount the pipette and cut the tip of the injection pipette to prevent blockage.
- Inject using the Nanoject until a droplet of mineral oil forms at the tip.
- Dilute virus for injection to desired concentration using saline.

Preparing the animal

- Weigh the animal.
- Anaesthetise the animal.
- Apply eye ointment .
- Buprenorphine/meloxicam subcutaneous injection (body weight x10 in μl using insulin syringe).
- Shave the animal.
- Place on stereotax.
- Drape the animal with press&seal.

Scrub up to sterile

- Wash hands.
- Put on sterile surgical gloves.
- Adjust drape so that continuous sterile field without any gaps is formed.
- Cut out a window in the drape about the incision site – ideally there should be no unshaved/nondisinfected areas in the window.
- Place sterile foil on all nonsterile touch points + drill.
- Mount drillpiece into the drill.

Implantation surgery

- Cut the skin using scalpel.
- Clean wound with cotton buds and PBS.
- Position stereotax pipette to Bregma. Zero x and y coordinates.
- Position stereotax pipette to position of injection (e.g. MPOA Bregma 0.00, 0.5 lateral). Mark the injection spot using a insulin syringe.
- Drill a hole in the skull. Gently enlarge the hole.

- Load diluted virus. Expel loaded air.
- Move over the injection pipette to the marked injection site and inject the loaded virus.
- Remove pipette slowly (every digit should be seen when the pipette is moving out of the brain).
- Position adapter on the stereotax, secure a syringe with 27 G needle. Lower the needle next to the viral injection site into the brain to the depth of GRIN lens implantation. This is to create a track for implantation
- Gently use the implantation rod to magnetically secure the GRIN lens at the end of the rod. Replace the syringe on adapter with implantation rod.
- Gently lower the implantation rod to start implantation into the brain. Perform up and down movement when implanting (0.1mm down and 0.05mm up) until the lens reach the depth of implantation.
- Use uv glue to create a small cone to secure the implant on the skull. Disconnect the GRIN lens implant from the implantation rod.
- Further enlarge the cone using superbond glue. After finishing gluing the implant, gently lift the skin to cover the cone.
- Suture up the wound.